

NADIM MAHMUD

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PROFESSIONAL SUMMARY

Python Backend Developer with hands-on experience building AI-powered backend services, integrating third-party and internal AI microservices, and delivering scalable production systems. Proficient in Django, Django REST Framework, PostgreSQL, and Docker, with strong experience in asynchronous programming, Celery/Redis/RabbitMQ background task processing, and WebSocket-based real-time systems. Experienced collaborating with cross-functional engineering teams on complex, multi-role platforms, with growing interest in fintech and financial technology applications. Strong analytical foundation with a solid grasp of Object-Oriented Programming (OOP), actively advancing Data Structures and Algorithms (DSA) skills, and backed by 600+ competitive programming solutions.

TECHNICAL SKILLS

- **Languages:** Python, JavaScript
- **Frameworks:** Django, DRF, React, Redux
- **Frontend & Styling:** HTML, CSS, Tailwind, Bootstrap
- **Databases:** PostgreSQL, MySQL, SQLite
- **Brokers & Cache:** Redis, RabbitMQ, Celery
- **Backend & Architecture:** RESTful API, WebSocket, JWT & RBAC, API Caching, Throttling, Asynchronous Programming, AI Microservices, System Optimization
- **Tools & DevOps:** Git, Docker, VPS, AWS, CI/CD, Postman

PROFESSIONAL EXPERIENCE

Jr. Backend Developer | *Join Venture AI – Dhaka, Bangladesh*

Oct 2025 – Present

- Designed, developed, and maintained scalable RESTful backend services using Python and Django REST Framework for AI-powered business applications.
- Integrated third-party APIs and internal AI microservices; implemented JWT-based authentication and role-based access control for secure, production-ready systems.
- Optimized system performance, scalability, and reliability through PostgreSQL query tuning, Redis-based API caching, efficient schema design, and implementing rate limiting/throttling.
- Built reliable, non-blocking backend workflows using asynchronous programming and Celery/Redis background task processing, integrated with machine learning pipelines.
- Wrote and maintained comprehensive unit and integration tests to ensure the stability and reliability of REST APIs and backend processes.
- Collaborated with product and engineering teams using Docker and AWS-based cloud deployment workflows, following clean architecture and scalable coding practices.

Team Leader | *Vcube Soft and Tech Limited – Dhaka, Bangladesh*

Jun 2019 – Oct 2024

- Promoted from Operator to Project Quality Checker, Mentor, and Team Leader through consistent performance.
- Managed team coordination, quality assurance, and workflow optimization for 2D/3D building floor plan projects.
- Mentored new team members and maintained productive communication across remote and in-office teams.

PROJECTS

AI Powered Insurance App

- Built a scalable insurance management platform featuring a freemium model: free-tier users interact with a real-time AI chatbot via WebSocket for insurance guidance, while premium subscribers access full case management with document submission and AI-driven, document-aware guidance.
- Managed WebSocket session flow and routed structured conversation data to an internal AI microservice for context-aware chat responses; handled connection management and session-based conversation state.
- Implemented manual email-based registration, Firebase-powered social authentication (Google, Apple), in-app purchase subscription system, role-based access control, JWT-based authentication with API throttling, cursor-based pagination for chat history, and Celery/Redis background tasks for automated weekly insurance claim reminders via Firebase Push Notifications.

Eat At Home

- Meal tracking and nutrition analytics platform; integrated Plaid API to fetch real bank transaction history for accurate restaurant spending tracking and automated food expense categorization.

- Called AI microservice to analyze food photos and return pricing estimates; implemented Redis caching for AI responses to reduce latency, and integrated results into weekly meal planning logic and Celery-based automated scheduling.
- Implemented manual email-based registration alongside Firebase-powered social authentication (Google, Facebook, Apple); built scalable backend using Django ORM, class-based views, serializers, and routers, using PostgreSQL for analytics data storage. Integrated WebSockets for live updates, Redis for background tasks, and Firebase for real-time push notifications.

RnD Taxbot

- Tax rebate automation platform with AI-powered WebSocket interview flow and multi-role approval workflow: user submission → admin-triggered AI scoring → approve or reject with resubmission.
- Orchestrated multi-turn AI interview sessions via WebSocket; designed secure document access control, and optimized PostgreSQL schema for complex multi-stage approval state management. Managed background workflows via Redis.
- Implemented admin-managed user provisioning, role-based access control, and JWT-based authentication across multi-role approval states.

Animal Service

- Full-featured pet marketplace with buyer/seller listings, auction system, vet appointment scheduling (online and offline), prescription generation, and vaccination record management; implemented Redis caching for high-traffic marketplace queries.
- Implemented proximity-based transport matching via Google Maps API — users search within a defined radius, nearby drivers receive real-time push notifications (integrated via Firebase and OneSignal), and a single driver accepts the request.
- Integrated Telr payments, digital wallet for seller earnings/withdrawals, and built real-time 1-on-1 chat using WebSocket; managed asynchronous background processes using Redis. Admin panel included with fraud prevention, verification workflows, and optimized pagination for large datasets.
- Implemented multi-method registration (email and phone), role-based access control for buyers, sellers, admins, veterinary doctors, and transport providers; optimized PostgreSQL schema across marketplace, bookings, medical records, and transactional workflows.

EDUCATION

Bangladesh Sweden Polytechnic Institute

Diploma in Computer Technology | Rangamati, Bangladesh

Jan 2014 – Jun 2018

KEY ACHIEVEMENTS

- Solved 600+ programming problems across Codeforces, LeetCode, HackerRank, CodeChef, and InterviewBit.
- Secured 38th place in the weekly CodeChef contest by solving all problems in June 2025.
- Achieved 11th position in the Inter Polytechnic Programming Contest at IUBAT.